



Comparison of zwitterions in amphiphilic PDMS-based surface-modifying polymers for marine fouling-release coatings

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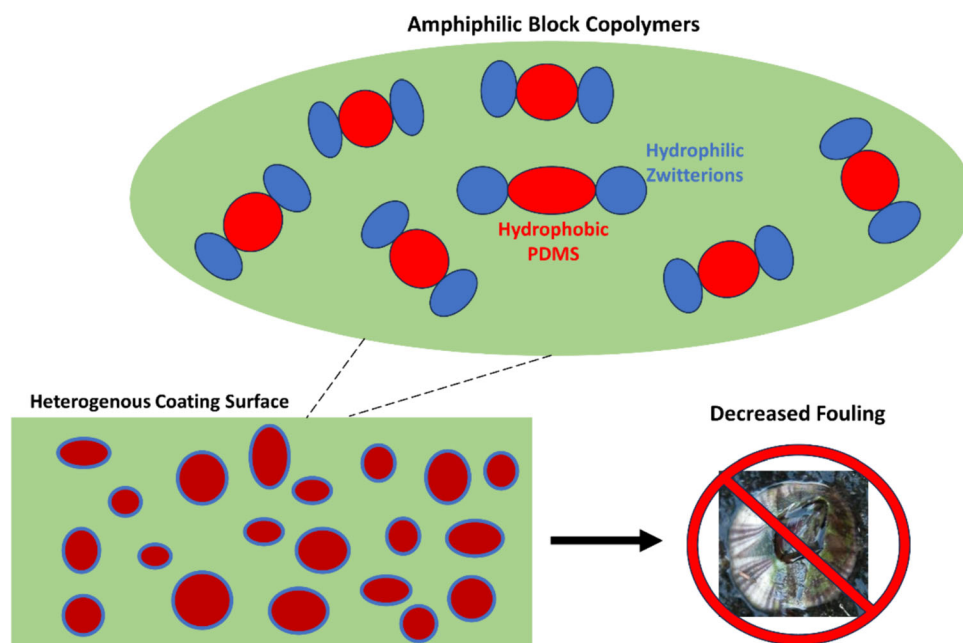
Abstract Zwitterion-containing polymers can be used as the hydrophilic block in amphiphilic copolymers due to their strong electrostatic interaction with water. Polyethylene glycol (PEG) is commonly used as the hydrophilic block, but PEG experiences rapid autoxidation which limits its lifetime. Alternatively, zwitterions are protein-resistant and have an ultra-low fouling surface that can bind water molecules stronger than PEG. In this study, the successful synthesis of block copolymers of hydrophobic poly(dimethyl siloxane) (PDMS) and four different hydrophilic zwitterionic monomers was carried out. The monomers were also successfully homo-polymerized. The eight polymers based on sulfobetaine methacrylate (SBMA), sulfobetaine methacrylamide (SBMAm), methacrylate phosphocholine (MPC), or carboxybetaine methacrylate (CBMA) were incorporated as a surface-modify-

ing additive into a polyurethane (PU) coating system to investigate the fouling resistance and fouling-release properties of the surfaces. The goal of this study was to determine which zwitterion copolymerized with PDMS produces the best-performing surface for FR purposes. Antifouling (AF) and fouling-release properties were examined through laboratory assays of three common marine fouling organisms: bacteria, algae, and barnacles. Detailed surface characterization was also conducted to investigate the heterogeneity of the surfaces. It was determined that the coatings containing the zwitterionic polymers improved the FR/AF properties compared to the control. The SBMA and CBMA zwitterionic polymers appeared to improve properties the most. For example, the incorporation of the PDMS-SBMA triblock copolymer reduced the barnacle adhesion from 0.58 to 0.19 MPa.

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Graphical Abstract



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Introduction

Marine biological fouling (biofouling) is the unwanted accumulation of marine organisms on artificial surfaces immersed in ocean water. The most common biofouling organisms on the microscale include bacteria, diatoms, and algae, while the most common on the macroscale are tubeworms, barnacles, and mussels.^{1,2} There are over 4000 discovered marine organisms that are considered foulers, and they have a large variety of sizes, affinities to different surfaces, and adhesion mechanisms. This phenomenon becomes even more complex with the process being multistaged and nonlinear. Seconds after the surface is immersed, a conditioning film of proteins and polysaccharides allows for the primary colonization of bacteria and microorganisms that form a biofilm/slime layer. This slime layer provides the basis for secondary and tertiary colonization for larger organisms to adhere to the surface.³ However, zoospores (*U. linza*) and some species of barnacles (*A. amphitrite*) can settle on pristine surfaces.^{4,5}

Biofouling has many negative consequences, besides the unappealing aesthetic effects, that include economic, safety-related, and environmental effects. The constant accumulation of biofouling on a ship's surface increases the frictional drag/resistance by increasing the roughness of the hull. This increased roughness causes increased fuel consumption, increased greenhouse gas emissions, and the spread of invasive

species.^{3,6,7} Estimates have shown that heavy calcareous fouling can result in powering penalties of 86% at cruising speeds. Even a light slime can cause a power penalty of 11% at cruising speeds.³ To address these penalties, the US Navy is projected to spend over \$1 billion a year on biofouling-related issues.⁴

The issue of biofouling has been a persistent problem throughout shipping history. Early protection of ship hulls incorporated lead and copper sheaths, but as ship hulls went from wood to steel, biocide-containing antifouling paints were introduced in the mid-19th century.¹ Many innovative attempts were made throughout the years to improve the antifouling properties when in the 1960s, tributyl tin (TBT) was introduced. The effectiveness of TBT was unmatched, but it had toxicity concerns for nonintended organisms and human health.⁸ Due to the adverse effects of TBT, it was banned worldwide and phased out by the International Marine Organization (IMO) in 2008.^{3,8} To address these new regulations, the focus has shifted toward less toxic biocides in antifouling (AF) coatings and fouling-release (FR) coatings to combat biofouling.

Antifouling coatings incorporate various biocides and boosters that deter the organisms from settling on the surface. Current AF coatings include copper-based biocides or other metal-based complexes.⁹ These biocides appear to be less toxic than TBT, but there are still toxicity concerns with these biocides and possible buildup in the environment.¹⁰ Fouling-release coatings do not typically contain any biocides or harmful chemicals that leach into the environment, making them more environmentally friendly. FR coatings function by weakening the adhesion between the

organism and substrate that facilitates easy removal when exposed to the hydrodynamic shear, such as a boat moving through the ocean or light cleaning/grooming.^{3,11,12}

Fouling-release coating systems have traditionally been made of elastomeric materials such as polydimethylsiloxane (PDMS) or fluoropolymers. These materials possess low surface energy and are hydrophobic, making them good candidates for weakening the adhesion of fouling organisms needed to be an FR coating.³ While much work is being done to improve the properties of silicone elastomer-based FR coatings,^{13,14} these materials are typically not very tough and suffer from weak adhesion to substrates, requiring a tie coat. To address these issues, a self-stratified siloxane-polyurethane (SiPU) coating has been developed.^{15–17} These coatings are tough and have good adhesion, removing the need for a tie coat. A limitation to implementing FR coatings is their lack of AF character when the ship is docked. As the ship is docked, fouling can accumulate and make it difficult to remove. Grooming/mechanical cleaning has been shown to maintain coatings free of fouling while in port, but a tough coating like the SiPU is needed to avoid damage.¹² Another downside to these hydrophobic, low surface energy coatings, is that some organisms (e.g., *N. incerta*) will preferentially adhere to hydrophobic coatings.^{3,11,17} Introducing amphiphilic polymers that include a hydrophobic and hydrophilic portion helps make the coatings effective across a wide array of organisms.^{12,18–21}

Amphiphilic polymers contain both a hydrophobic and hydrophilic portion that can incorporate antifouling character to a coating through the protein-resistant hydrophilic block of the polymer.^{22–24} Polyethylene glycol (PEG) is regularly incorporated in these amphiphilic polymers as it is nontoxic and can resist protein absorption on the surface of the coating. The hydrophilic interactions with water and PEG restrict protein absorption.²⁵ PEG integrated into an amphiphilic fouling-release coating has shown increased performance against marine organisms while also broadening the effectiveness of the coating.^{18–21} PEG can be a useful hydrophilic polymer but suffers from a decreased service life/long-term stability due to PEG experiencing rapid autoxidation due to transition metals present in the marine environment.²⁶ Other promising hydrophilic polymers that could be used to replace PEG are zwitterionic-based. Zwitterionic polymers are a promising candidate due to their hydrophilicity and ultra-low fouling pressure/protein resistance.^{27–31} Due to zwitterions having a negative and positive charge incorporated on the same molecule, binding with water is much stronger than PEG. This strong water-binding effect is due to the electrostatic interaction of zwitterions compared to the hydrogen bonding of PEG or other nonzwitterion materials.³² For example, a common zwitterion, poly(sulfobetaine methacrylate) (SBMA), will bind water up to 8× more than PEG.³³ This creates a very strong

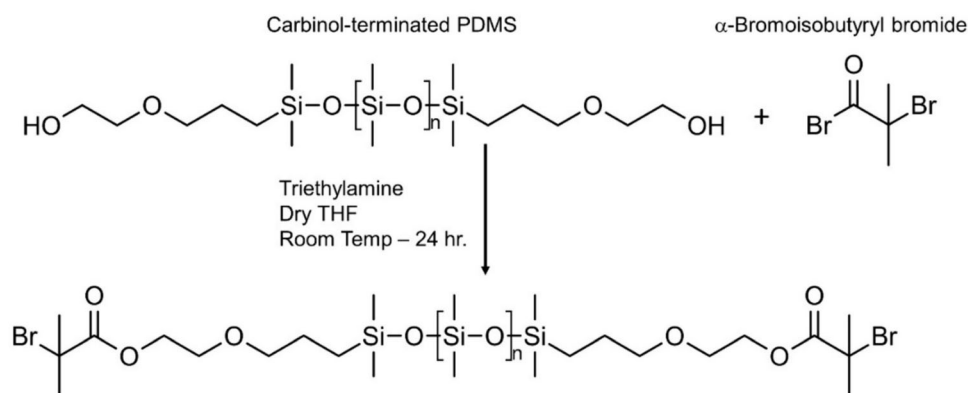
hydration layer that will not readily be displaced by marine organisms to settle on the surface of the coating. Another positive aspect of zwitterionic polymers over PEG is the increased long-term stability.²⁹ Zwitterionic materials such as sulfobetaine and carboxybetaine have been explored because they have shown increased resistance toward fouling on surfaces for the marine environment.^{28,34–38} Due to their increased stability and ultra-low fouling characteristics, zwitterions are a promising candidate for incorporation in fouling-release coatings.

The goal of this work was to compare the effects of four different zwitterionic monomers when synthesized as novel amphiphilic block copolymers with PDMS that has constant molecular weights. The homopolymers of each zwitterion were also investigated. These unique polymers were incorporated as additives into a polyurethane coating system at a constant concentration to tune the antifouling and fouling-release properties and determine which zwitterionic-based polymer would produce the best results for use in the future FR coatings. In this work, we report the successful synthesis of novel amphiphilic block copolymers with highly incompatible blocks of hydrophilic zwitterions and hydrophobic PDMS with a controlled molecular weight and polydispersity index (PDI) via a controlled free radical polymerization. ARGET-ATRP (activators regenerated by electron transfer atom transfer radical polymerization) was utilized to perform the synthesis because it provided control of the molecular weight without the need for a glovebox, and only low amounts of copper catalyst are required (<300 ppm). These amphiphilic polymers are designed to modify the surface properties, so the PDMS block was utilized to provide a hydrophobic block as well as facilitate stratification to the surface. The incompatibility of the PU (polar & high surface energy) and PDMS (nonpolar & low surface energy) allowed for stratification of the copolymer to the surface.^{11,15,17} Meanwhile, the zwitterionic block provides a hydration layer on the surface when immersed in water making it difficult for organisms to displace and adhere strongly. The amphiphilic polymer allows for a stratified surface-modifying additive to be incorporated into a PU coating to give it AF/FR properties. This study set out to fully characterize and compare the amphiphilic surfaces to determine how each zwitterion affects the surface properties as well as the biological AF/FR properties.

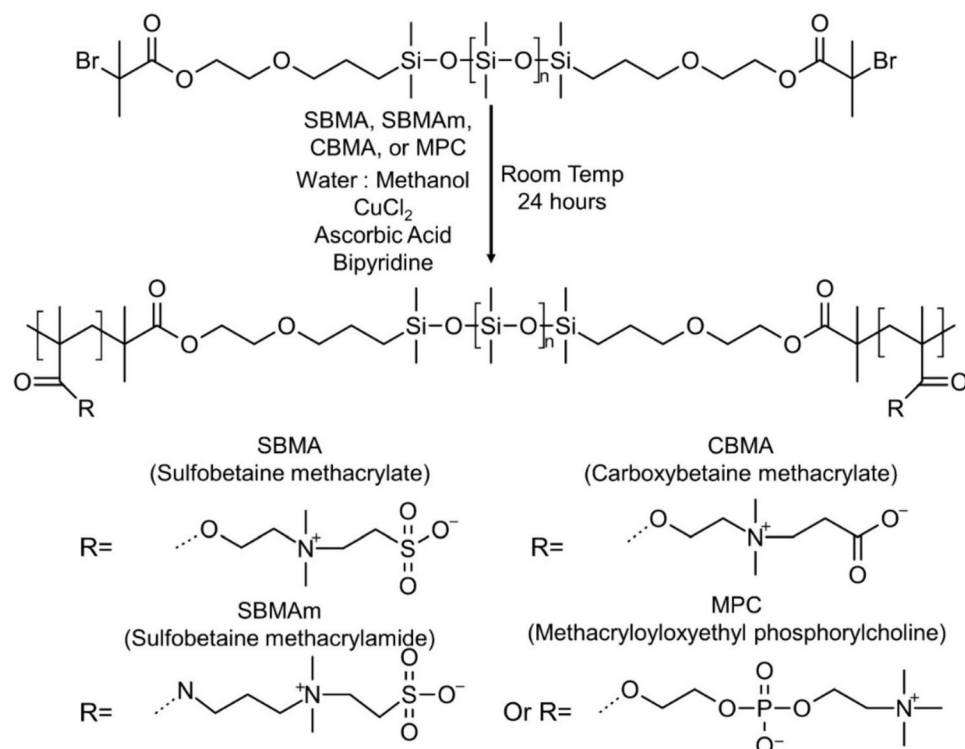
Experimental

Materials

Isophorone diisocyanate (IPDI) polyisocyanate Desmodur Z4470 BA was provided by Covestro LLC. Difunctional carbinol (hydroxy)-terminated polydimethylsiloxane (PDMS) of molecular weight 1000



Scheme 1: Synthesis of Br-terminated PDMS macroinitiator for ARGET-ATRP



Scheme 2: Synthesis of triblock amphiphilic copolymer additives via the ARGET-ATRP method

$\overline{M}_n$ (DMS-C15) was purchased from Gelest, Inc. Copper (II) chloride, ascorbic acid (vitamin C), 2,2'-bipyridine, α -bromoisobutyryl bromide (BIBB) 98%, triethylamine, ethyl- α -bromoisobutyrate 98%, dibutyltin diacetate (DBTDAc), 2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl) ammonium hydroxide (SBMA) 95%, N-(3-sulfopropyl)-N-methacryloylamidopropyl-N, N-dimethylammonium betaine (SBMAm) 95%, 2-methacryloyloxyethyl phosphorylcholine (MPC) 97%, 3-((2-(methacryloyloxy)ethyl) dimethylammonio) propanoate (CBMA) 98%, butyl methacrylate (BMA), tetrahydrofuran (THF), toluene, dichloromethane, magnesium sulfate, acetone, acetylacetone, methanol, isopropyl alcohol

(IPA), N-methyl-2-pyrindone (NMP), methyl amyl ketone (MAK), and drying molecular sieves of 4 Å size were purchased from Sigma-Aldrich. A solution of DBTDAc was prepared by mixing 1 wt.% in MAK. An acrylic polyol composed of 80% butyl acrylate, and 20% 2-hydroxyethyl acrylate was synthesized via conventional free radical polymerization and diluted to 50% in toluene.³⁹

AkzoNobel, International Paint Ltd. produced the commercial fouling-release standards Intersleek® 700 (IS 700), Intersleek® 900 (IS 900), and Intersleek® 1100SR (IS 1100). The silicone elastomer, Silastic® T2 (T2), was provided by Dow Corning. Aluminum panels (4'' × 8'', 0.6-mm-thick type A, alloy 3003 H14) were

Table 1: Experimental polyurethane coatings formulations made with 5 wt.% zwitterionic polymer

Formulation	Details
F0—control	Polyurethane control (No additive)
F1—SBMA-PDMS	SBMA-PDMS-SBMA triblock polymer
F2—Poly(SBMA)	SBMA homopolymer
F3—SBMAm-PDMS	SBMAm-PDMS-SBMAm triblock polymer
F4—Poly(SBMAm)	SBMAm homopolymer
F5—MPC-PDMS	MPC-PDMS-MPC triblock polymer
F6—Poly(MPC)	MPC homopolymer
F7—CBMA-PDMS	CBMA-PDMS-CBMA triblock polymer
F8—Poly(CBMA)	CBMA homopolymer

purchased from Q-lab. The aluminum panels were sandblasted and primed with Intergard 264 (International Paint) using air-assisted spray application. Multiwell plates were modified using circular disks (1-inch diameter) of coatings on primed aluminum.⁴⁰

Synthesis of ARGET-ATRP macroinitiator (bromine-terminated PDMS)

The reaction that was used to synthesize the Br-terminated PDMS macroinitiator is shown in Scheme 1. The molar ratio of hydroxy-terminated PDMS:triethylamine:BIBB was 1.00:2.00:2.32. A detailed description of this reaction can be found elsewhere.³⁶ Briefly, in a 500 mL round-bottom flask, carbinol-terminated PDMS (30.000 g) and triethyl amine (6.071 g) were dissolved in 200 mL dry THF. BIBB (16.001 g) was added dropwise to the flask at a temperature of 0–5°C. After addition, the reaction was stirred overnight, a white precipitate was removed with a fritted funnel, and the product was washed with dichloromethane. The synthesized product was characterized via Fourier transform infrared spectroscopy (FTIR) and gel permeation chromatography (GPC).

Synthesis of zwitterionic-based copolymers

The amphiphilic polymers that contained blocks of zwitterions (SBMA, SBMAm, MPC, or CBMA) and PDMS were synthesized through the ARGET-ATRP process. This is shown in Scheme 2. The ratio of [monomer]:[macroinitiator]:[catalyst]:[reducing agent]:[ligand] for the polymerization was 100:0.2:0.01:0.15:0.05. All of the synthesized triblock copolymers had a constant molecular weight of 3000 $\overline{M}_w$ (1000 $\overline{M}_w$ for each zwitterion block (2000 $\overline{M}_w$ total) and 1000 $\overline{M}_w$ for PDMS block). For example, to synthesize 3 g of triblock CBMA-based additive with each block at 1000 $\overline{M}_w$ (CBMA-PDMS-CBMA), CBMA monomer (3000 mg), macroinitiator PDMS (2372 mg), copper chloride catalyst (12 mg), ascorbic

acid reducing agent (242 mg), and bipyridine ligand (71 mg) were added together. A detailed description of synthesis using SBMA monomer can be found elsewhere.³⁶ The synthesized triblock amphiphilic additives were characterized via FTIR. As GPC could not be run on these copolymers due to their insolubility in common organic solvents (THF), a test synthesis to confirm the target molecular weight using GPC was done with butyl methacrylate, and a targeted 3000 $\overline{M}_w$ polymer was confirmed. The respective homopolymer with targeted 5000 $\overline{M}_w$ for each zwitterion was also synthesized using the same procedure using ethyl- α -bromoisobutyrate initiator. All homopolymers had a constant 5000 $\overline{M}_w$ molecular weight.

Coating formulations

All eight synthesized additives were added to a polyurethane formulation at a constant 5 wt.% concentration to determine and compare the effects of the different amphiphilic zwitterions and their respective homopolymers on coating performance. The polyurethane coating system consisted of an acrylic polyol crosslinked with an isocyanate trimer resin (IPDI). The only difference between the coating formulations was the additive incorporated; the molecular weight of additives was all constant and so was the amount. NMP and IPA solvent were added to help dissolve the additives in the formulations. Table 1 shows all of the experimental formulations. For example, to formulate a coating with 5 wt.% of CBMA triblock additive (F7), additive (0.83 g), NMP (2.02 g), and IPA (0.98 g) were added to the vial and mixed in a vortex mixer for about 15 min until the additive was mostly dissolved in the solution. Acrylic polyol with 50% solids (23.12 g) and acetone (1.6594 g) were added to the vial and magnetically stirred at room temperature for 24 h. IPDI trimer resin Desmodur Z4470BA (7.46 g) and DBTDAC catalyst solution (0.83 g) were added to the vial, and the mixture was stirred at room temperature for 1 h. After mixing, coating formulations were drawn down on primed 8" × 4" aluminum panels with a wire-

Table 2: Control and standard coatings list

Coating	Coating ID	Description
Polyurethane	F0	Control PU with no additive
Commercial PU	PU	Pure polyurethane standard
Dow ® T2	T2	Silicone elastomer standard
Intersleek ® 700	IS 700	Intersleek ® silicone elastomer standard
Intersleek ® 900	IS 900	Intersleek ® amphiphilic silicone elastomer standard
Intersleek ® 1100SR	IS 1100	Intersleek ® amphiphilic silicone elastomer with slime release technology

round drawdown bar with a wet film thickness of 80 μm . All coatings were cured under ambient conditions for 24 h followed by oven curing at 80°C for 45 min. Fully cured panels had disks punched out and put into well plates for laboratory biological assays.

Experimental design

This study was designed to assess eight additives based on four different commercially available zwitterions. Four of the additives were triblock copolymers with PDMS, and four were homopolymers of the zwitterions. To compare the effect of just the composition of the zwitterion on the fouling-release performance, the molecular weight of all the blocks was kept the same (1000 $\overline{M}_w$ for triblock and 5000 $\overline{M}_w$ for homopolymer), and concentration (5 wt.%) of additive in the coating was kept constant. Additives were incorporated into a polyurethane matrix system to study the effect the additives had on the system and how they self-stratified on the surface without interference from any other moieties, e.g., siloxane polyurethane systems.

Control and standard coatings

Commercial standards were prepared following the manufacturer's specifications. A base polyurethane coating was used as the control. Similar to the experimental formulations, controls, and standards were prepared on 4" $\times$ 8" primed aluminum panels and disks cut out for well plates. Table 2 contains detailed descriptions of controls and standards.

Fourier transform infrared spectroscopy

Fourier transform infrared (FTIR) spectroscopy was used to characterize the additives, using a Thermo Scientific Nicolet 8700 FTIR. The additive was applied as a thin layer on a potassium bromide (KBr) plate to collect the spectrum.

Surface characterization

Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) was utilized to characterize the surface of the coatings and confirm the presence of surface-modifying additives. A Thermo Scientific Nicolet 8700 FT-IR spectrometer, with a Smart iTR diamond plate ATR sampling accessory, was used to collect ATR-FTIR spectra of coatings.

A Kruss® DSA (Drop Shape Analyzer) was used to measure the surface wettability and to calculate the surface energy of the coatings. Five replicate water and diiodomethane contact angles were measured for each sample. The static contact angles were measured over 10 min. Surface energy was calculated via the Owens–Wendt method.⁴¹ Five replicates were recorded, and data represent the representing mean values with error bars. The angles and surface energies were calculated using Kruss® Advance software.

Atomic force microscopy (AFM) was incorporated to investigate the surface topography of the coatings with surface-modifying additives. A Jupiter XR Atomic Force Microscope from Asylum Research was used to scan the surface of the coatings, collecting images on a sample area of 100 μm $\times$ 100 μm in tapping mode. The experiments were run in air and deionized water under ambient conditions, using a silicon probe with a constant spring constant of 0.1–0.6 N/m and resonant frequency of 15–39 kHz. For each surface, at least three different sites at varying spots on the coatings were scanned to ensure representative images were obtained.

Photoinduced force microscopy (PiFM) was utilized to characterize the surface topography and composition of the coatings. A Vista One photoinduced force microscope for nano-IR chemical analysis from Molecular Vista scanned the surface of the coatings, collecting images on a sample area of 26 μm $\times$ 26 μm in noncontact mode. The experiments were run in air, under ambient conditions, using platinum–iridium (PtIr) cantilevers with a spring constant of 10–130 N/m and resonant frequency of 204–497 kHz. For each sample, three replicates at varying spots were collected to ensure accuracy and consistency.

X-ray photoelectron spectroscopy (XPS) was conducted using a Thermo Scientific™ K-Alpha™ system to determine the elemental composition of the coatings at the surface and below the surface. The instrument is

equipped with a monochromatic Al K_α (1486.68 eV) X-ray source and Ar⁺ ion source (up to 4000 eV). Depth profiling of a coating was evaluated using argon-ion with 30 etch cycles. For each etching cycle, the ion beam was set to 1000 eV monatomic mode with low current and 30 s etch time. After each etching cycle, five replicate survey spectra were collected, at low resolution, with a constant analyzer pass energy of 200 eV, for a total of 20 ms. For each run, photoemission lines for C1s, N1s, O1s, and Si2p were observed. Spectra were collected at an angle normal to the surface (90°) of a 400-μm area. The chamber pressure was maintained below 1.5×10^{-7} torr, and samples were analyzed at ambient temperature. Atomic concentrations were quantified by the instrument's software as a representation of the atomic intensities as a percentage of the total intensity of all elements.

Moisture adsorption of the coatings' surfaces was characterized by utilizing a VStar 4 Station Turbo from Anton Paar. VStar software from Anton Paar was used to analyze the data and create absorption curves at various pressures. The instrument is equipped with an onboard degassing apparatus where the coating samples were subjected to 100°C at 0.5 torr for 24 h to remove atmospheric moisture. The temperature control system maintained a 40°C environment for the coating samples, and the water vapor was maintained at 110°C for homogeneous vapor concentration. Adsorption points were measured at increasing pressures to create adsorption curves that could be used to compare samples' hydrophilicity/hydrophobicity. Samples were prepared by cutting an aluminum panel into 4 cm × 1 cm coupons and dip-coated so that all sides were completely covered. The samples were allowed to cure and then weighed before testing. After the degassing procedure, samples were reweighed to see the weight loss, and this new weight was entered into the software and used to calculate the water adsorption based on free volume measurements by the instrument.

Water aging

All the prepared coatings were preleached for 28 days in running tap water. The water tanks were equipped to automatically fill and empty every 4 h. Water aging of the coatings was carried out to meet two requirements. The first was to leach out any impurities that may be toxic and interfere with FR assessments. The second was to determine if there were any surface rearrangements of the coatings and whether the additives leached out to a significant degree. All biological assays were done after the preleaching process was completed.

Biological laboratory assays

Bacterial (Cellulophaga lytica) biofilm adhesion

Fouling-release properties toward bacteria were evaluated using retention and adhesion assays as described in more detail in the Supplementary Information (SI).^{40,42,43} Briefly, the marine bacterium was suspended in artificial seawater, fed, and incubated for 24 h. After this time, the number of organisms on the surface was measured and then the cells were pressure washed. The number of organisms remaining after the pressure washing was measured again and reported.

Growth and release of microalgae (Navicula incerta)

The laboratory biological assay with the diatom (*Navicula incerta*) was conducted following a similar procedure to that described in SI.^{5,44,45} Briefly, the algae were suspended in artificial seawater, fed, and incubated for 2 h. After this time, the number of organisms on the surface was measured and pressure washed. The number of organisms remaining after pressure washing were measured.

Adult barnacle 2-week reattachment and adhesion (A. amphitrite)

To assess marine barnacle (*A. amphitrite*) attachment and adhesion, a procedure outlined in the SI was performed.^{46,47} Briefly, adult barnacles were placed on coating panels in artificial seawater and fed for 2 weeks. After 2 weeks, the number of barnacles attached to the surface was recorded, and the stress needed to remove them from the coating was measured with a force gauge.

Statistical analysis

Statistical analyses were performed using JMP Pro software, version 17. The GLM procedure with Tukey's method was utilized to determine the difference in mean for each treatment group under a completely randomized design. The assessed response for this analysis was the biomass remaining of the marine organisms at 20 PSI for *N. incerta* and *C. lytica*. The force needed to remove barnacles after reattachment was the assessed response for *A. amphitrite*.

Results and discussion

Zwitterionic polymers have been employed as a hydrophilic component in amphiphilic systems to increase the antifouling (AF) and fouling-release (FR) properties of marine coatings. In this study, four

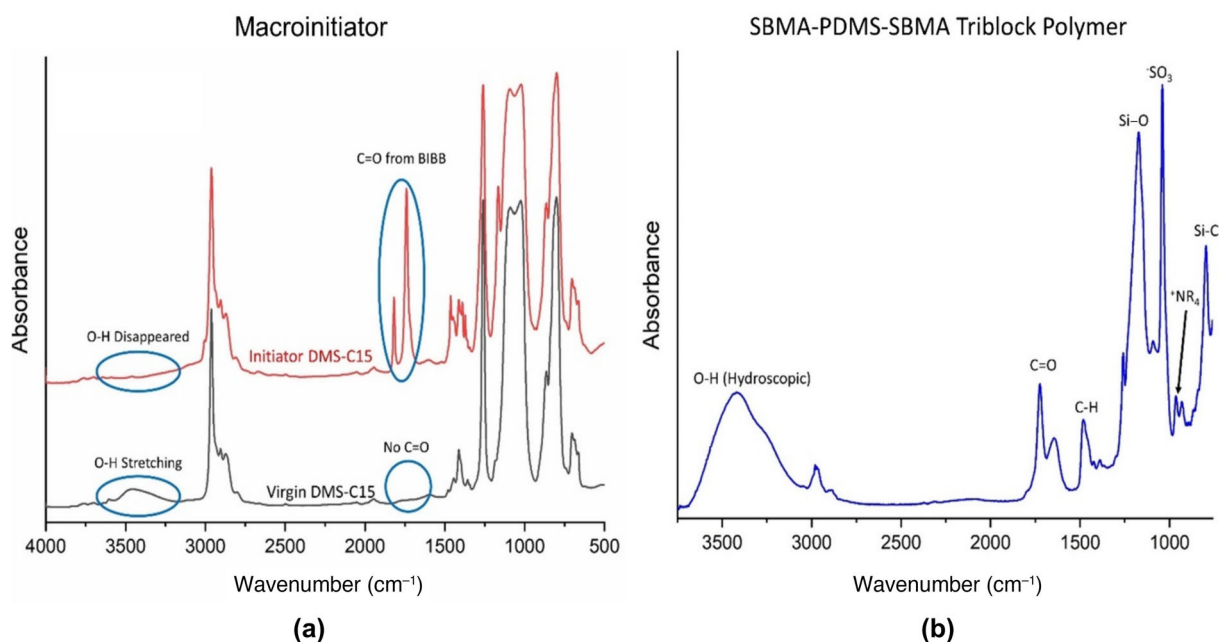


Fig. 1: Fourier transform infrared (FTIR) spectra of (a) OH-terminated PDMS and the synthesized Br-terminated PDMS macroinitiator and (b) SBMA-PDMS-SBMA triblock copolymer

commercially available zwitterionic monomers were polymerized with PDMS to create amphiphilic triblock polymers as well as their respective homopolymers to compare the different zwitterions' performance. The concentration and molecular weight of the polymers were kept the same to study the relationship between the zwitterions and determine which yielded the best AF/FR performance.

ARGET-ATRP macroinitiator synthesis

The amphiphilic polymer synthesis required two steps to complete. The first step involved reacting a carbinol-terminated PDMS of 1 K average molecular weight (DMS-C15) with BIBB. This results in a Br-terminated PDMS that will function as the ARGET-ATRP macroinitiator as outlined in Scheme 1. The transformation of the hydroxyl end groups to bromine is supported by the disappearance of the OH peak (3500 cm^{-1}) and the appearance of the C = O peak (1740 cm^{-1}) in the FTIR spectra shown in Fig. 1a. There was also a color change to a slight yellow as the BIBB reacted with the hydroxyl groups of the PDMS.

Zwitterionic polymer syntheses and characterization

ATGET-ATRP synthesis was used to make the triblock copolymers. This technique allows for the chain length/molecular weight of each block to be controlled and requires negligible catalyst concentrations (ppm scale). Another large advantage of AR-

GET-ATRP over traditional ATRP is that there is no need for a glovebox as the catalyst is relatively stable. ARGET-ATRP requires five components that include monomer (zwitterions), macroinitiator (Br-terminated PDMS), catalyst (copper (II) chloride), reducing agent (vitamin C), and ligand (2,2'-bipyridine).³⁶

Before the zwitterionic polymers were made, a test synthesis was done using butyl methacrylate since that copolymer is soluble in organic solvents and amenable to characterization. This allowed for GPC to be run to confirm that the correct molecular weights were obtained from the synthesis. FTIR of the butyl methacrylate triblock polymer with PDMS and its homopolymer are shown in Fig. S1 (Supplementary Information). The GPC results also confirmed that the triblock polymer had an $\overline{M}_w$ of 3240 and a polydispersity index (PDI) of 1.16. The homopolymer had an $\overline{M}_w$ of 5085 and PDI of 1.35. This confirmed a successful synthetic procedure, and the zwitterionic monomers can be polymerized.

Amphiphilic block ABA copolymers were made by reacting the synthesized macroinitiator and monomers together via a controlled free radical polymerization outlined in Scheme 2. The homopolymers with the four zwitterionic monomers were also synthesized following a similar method. The formation of the SBMA copolymer is confirmed by FTIR as shown in Fig. 1b. The data have inductive signals for both SBMA and PDMS: C = O (1740 cm^{-1}), Si-O (1160 cm^{-1}) SO_3^- (1060 cm^{-1}), $^+\text{NR}_4$ (960 cm^{-1}), and Si-C (750 cm^{-1}). The FTIR for the SBMA homopolymer is shown in Fig. S2. The spectra of the other six polymers are shown in Figs. S3, S4, and S5.

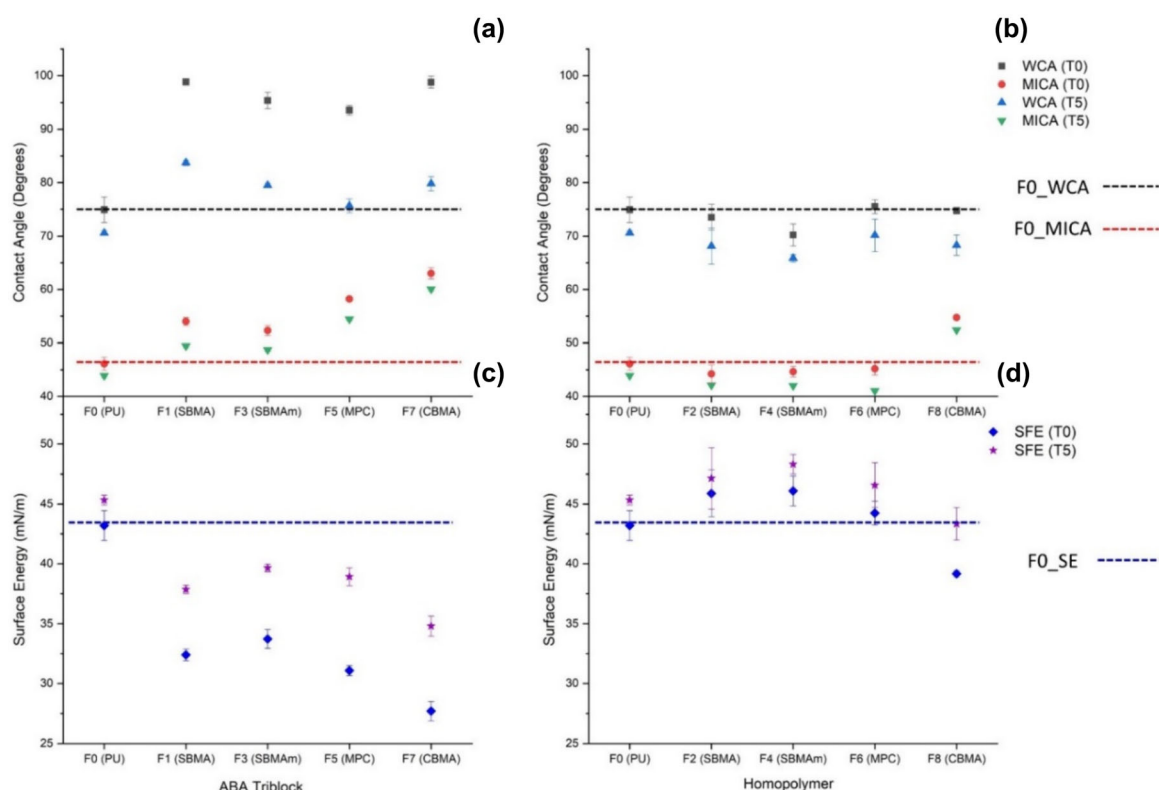


Fig. 2: (a) Water contact angle (WCA) and methylene iodide contact angles (MICA) for triblock coatings at T0 (0 min) and T5 (5 min); (b) WCA and MICA for homopolymer coatings at T0 (0 min) and T5 (5 min); (c) surface free energies (SFE) for triblock coatings; and (d) SFE for homopolymer coatings. Dashed lines are to guide the eye for easy comparison with F0 (polyurethane control) at T = 0 min

Surface characterization of coatings

Attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy can be utilized to determine the functional groups that are present near the surface. The estimated depth of penetration of the IR is hundreds of nanometers.⁴⁸ For most cases, this will give a general idea of how the surface and the material are comprised; however, if the material is self-stratified directly on the surface and is not distributed through the penetration depth, the FTIR will not show the self-stratified material. This appears to be the case for the zwitterion coatings that have a 5 wt.% zwitterion polymer/additive incorporated. The modified coatings show essentially the same spectra as the control coating. The indicative PDMS and zwitterion peaks do not appear, and the polyurethane C–O and C=O are seen. The ATR-FTIR spectra are shown in Fig. S6. As the concentration of the additive is not as high as it is on the surface, the additive signals are low and hidden under the polyurethane peaks. Other more surface-sensitive techniques, such as XPS and PiFM, are required in order to understand the surface compositions of the coatings.

Contact angle measurements were employed with a polar solvent, water, and a nonpolar solvent, methylene iodide, to characterize how the amphiphilic

surfaces behave as a function of time. The contact angle data were collected as static measurements at 0 min (T0) and 5 min (T5). Data for the triblock coatings are shown in Fig. 2a and the homopolymer coatings in Fig. 2b. The initial water contact angle is much higher than the control due to the presence of the PDMS, but the WCA decreases with time due to the zwitterionic nature also present on the copolymer. The large change in water contact angles over time for the triblock copolymer coatings when compared to control can be attributed to the amphiphilicity of the coating as the water droplet spreads out and tries to find equilibrium over time. The change in methylene iodide contact angle over time is not as significant due to the nonpolarity of the solvent, but some changes can still be seen. This observation supports the hypothesis that the triblock copolymer additive is self-stratified to the surface of the coating and results in a change in the surface properties. For the homopolymer coatings, the initial WCA is much lower than the triblock polymer coatings and is similar to the control coating. These coatings are much more hydrophilic than the amphiphilic coatings and because there is no amphiphilicity present, the change of contact angle over time is not as significant. Again, due to the polarity of the methylene iodide, the MICA changes are not too different from the control. This observation can help explain that the

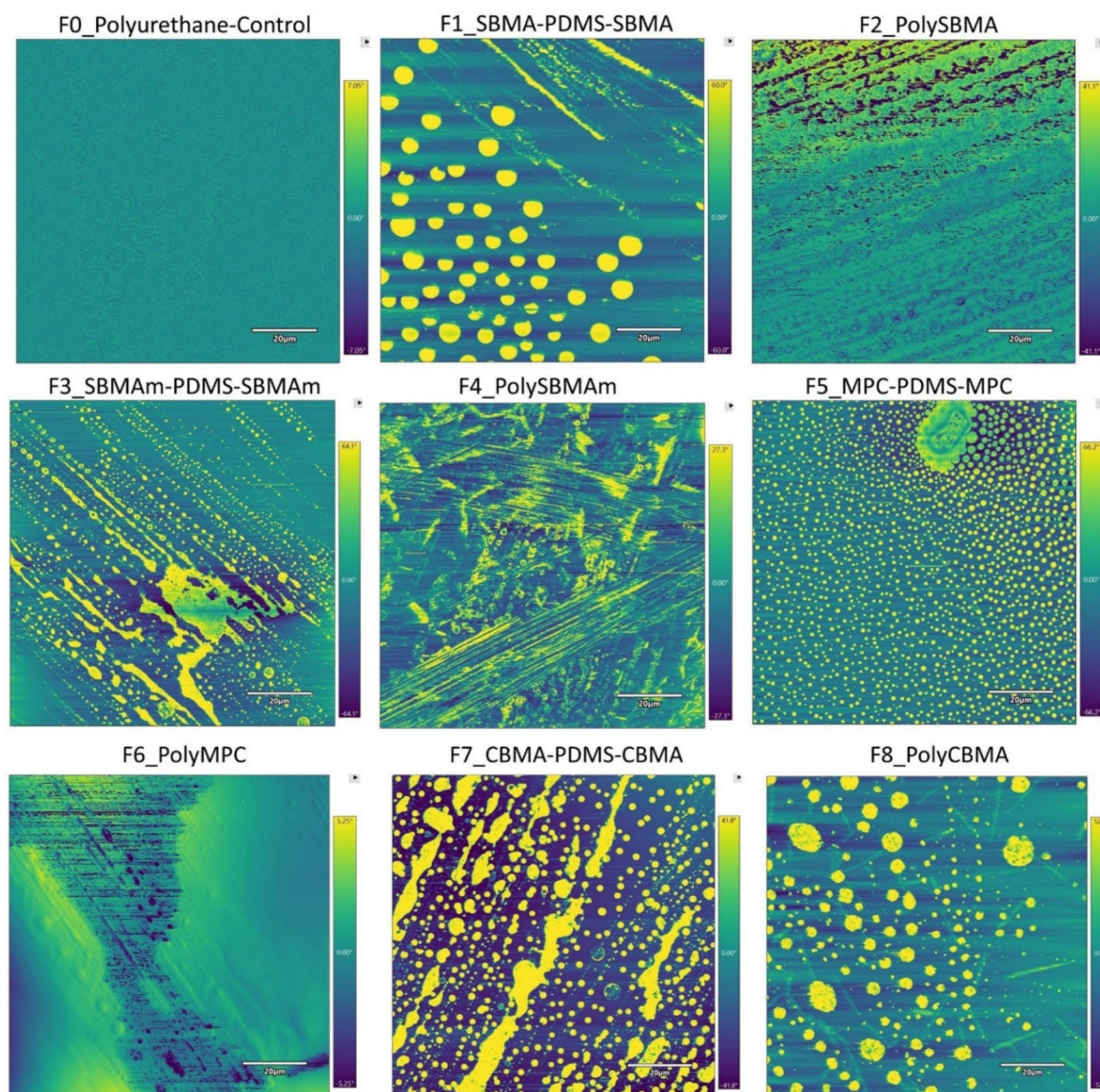


Fig. 3: AFM ambient/air phase images for formulations F0–F8. Each image is for an area of $100\ \mu\text{m} \times 100\ \mu\text{m}$

PDMS is needed to help the additive self-stratify to the surface to help modify the surface properties, as is observed from the contact angle data.

Surface energies were calculated from the WCA and MICA using the Owens–Wendt method. The surface free energies (SFE) for the triblock polymer coatings are shown in Fig. 2c and the homopolymer coatings in Fig. 2d. The initial SFE for the triblock coatings are in the range of 27–33 mN/m. Due to the PDMS present on the polymer, this is significantly lower than the control coating of 42 mN/m. However, because of the presence of the zwitterions, the SFE changes over time as the droplets spread out. The SFE after 5 min are in the range of 35–39 mN/m. Again, this observation can be correlated to explain that the amphiphilic additives are stratified to the surface and impact the surface energies. For the homopolymer coatings, the initial

SFE measurements are slightly higher and are similar to the control coating. Due to there not being any amphiphilic nature in these coatings, the surface free energies do not change much with time. There may be some impact from the zwitterions affecting the surface energies, but it is not significantly different than the control. This implies that the PDMS portion of the polymer could be vital in helping the additive stratify to the surface for modification of the surface properties.

A Keyence digital microscope was utilized to image the surface at a magnification of $20\times$ and $100\times$. This was done because some of the coating formulations, especially those made with the homopolymer zwitterions, had difficulty remaining fully mixed in the coating matrix as the coatings were cured. These images are shown in Figs. S7, S8, S9, and S10. The images show

Table 3: AFM roughness values for ambient and hydrated coatings

Formulation	Air		Hydrated	
	Rq (nm)	Ra (nm)	Rq (nm)	Ra (nm)
F0_PU	4	3	6	1
F1_SBMA-PDMS	53	35	109	70
F2_SBMA	54	39	77	45
F3_SBMAm-PDMS	65	39	166	137
F4-SBMAm	33	22	73	45
F5_MPC-PDMS	44	38	70	27
F6_MPC	77	64	60	45
F7_CBMA-PDMS	72	43	95	47
F8_CBMA	33	16	75	61

that when compared to the polyurethane control, there are domains that are present across the surface; however, these domains differ in size and vary based on the type of polymer (triblock vs. homopolymer) and type of zwitterion. This experiment shows that the homopolymer zwitterions may be too hydrophilic to completely solubilize in the polyurethane coating matrix, and the homopolymer appears to agglomerate in larger areas when compared to the triblock polymers.

Atomic force microscopy (AFM) was used to investigate the morphology of the zwitterionic-modified surfaces. The phase images in AFM correspond to the difference in materials as the tip interacts with these materials in different ways. Large phase differences will correspond to heterogeneous surfaces that have multiple materials present. The height images in AFM can also show how pronounced or depressed these domains are on the surface, like mountains and valleys. The modified coatings have dark and light patterns in the phase images suggesting a complex morphology at the surface (Fig. 3). The corresponding height images are shown in Fig. S11. The images were run both in ambient air conditions as well as hydrated in water. The hydrated phase images are shown in Fig. S12, and the corresponding hydrated height images are shown in Fig. S13. The roughness values of the height images in air and water are shown in Table 3. The unmodified F0 control displays a homogeneous surface without the presence of domains on the surface with little roughness, which is expected as there is nothing to modify the coating. As the zwitterionic additives are introduced into the system, the presence of heterogeneous domains on the surface becomes apparent. The amphiphilic zwitterionic triblock copolymer additives appear to have more spherical microdomains (odd formulations) while the homopolymer zwitterionic polymers (even formulations) have more streak-like surfaces. This can be attributed to the fact that the amphiphilic triblock copolymers are segregating to the surface and creating more spherical domains. On the other hand, the

zwitterionic homopolymers are more difficult to disperse in the resin matrix and are coming to the surface to create these complex heterogeneous domains. It can be noted that the type of polymer additive (amphiphilic versus homopolymer) and type of zwitterion can be seen to play a role in the surface appearance and some of the surface properties. When comparing the roughness values, two things can be concluded: The incorporation of the polymer/additive creates a rougher surface and when the coatings are immersed in water, the roughness increases. These observations can be attributed to the additives/polymers segregating to the surface as well as these extremely hydrophilic polymers swelling under the presence of water. This swelling of the polymers under water confirms the use of these additives in helping to create a hydration layer that makes it difficult for fouling organisms to displace and adhere to the surface.

Photoinduced force microscopy (PiFM) was utilized in conjunction with AFM to determine the composition of the microdomains and to obtain compositional analysis of the surfaces. PiFM allows the user to obtain nano-IR spectra at specified points on a noncontact AFM image as well as take full images at specified IR wavelengths. The instrument has two modes of collecting these IR spectra and images. Sideband mode is more sensitive and suppresses background signals that are extremely surface sensitive with a laser penetration depth of < 25 nm. Direct mode (“bulk”), the other imaging technique, modulates light at the resonant frequency of the cantilever vibration mode and takes spectra around 100 nm into the coating.^{49–51} With these two modes of imaging the coatings, a clear picture of what is directly on the surface and just below the surface can be obtained. The PiFM image and infrared spectra taken at different locations for formulation F7 (CBMA-PDMS-CBMA) is shown in Fig. 4. The PiFM images for the other formulations are found in Figs. S14, S15, S16, S17, S18, S19, S20, and S21. After taking a noncontact AFM image, the probe can be pinpointed inside the spherical domains and outside these domains. When imaging in the surface-sensitive

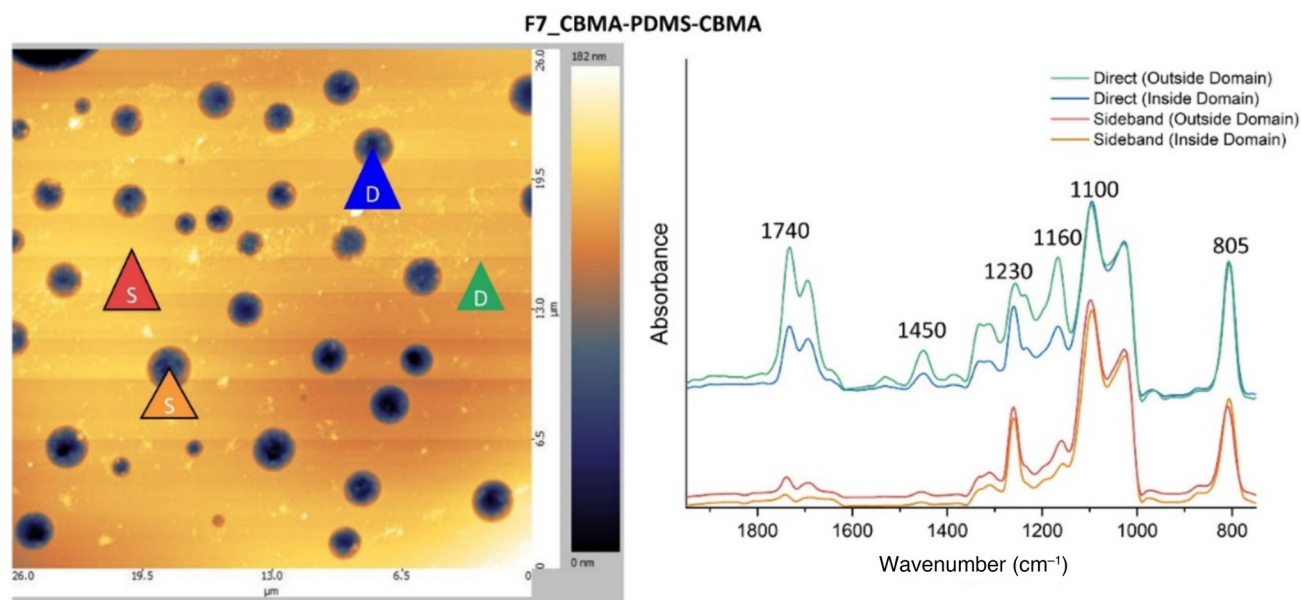


Fig. 4: PiFM image of formulation F7 (CBMA-PDMS triblock) with spectra inside and outside domains. Sideband mode measures < 25 nm depth and direct mode measures ~ 100 nm depth

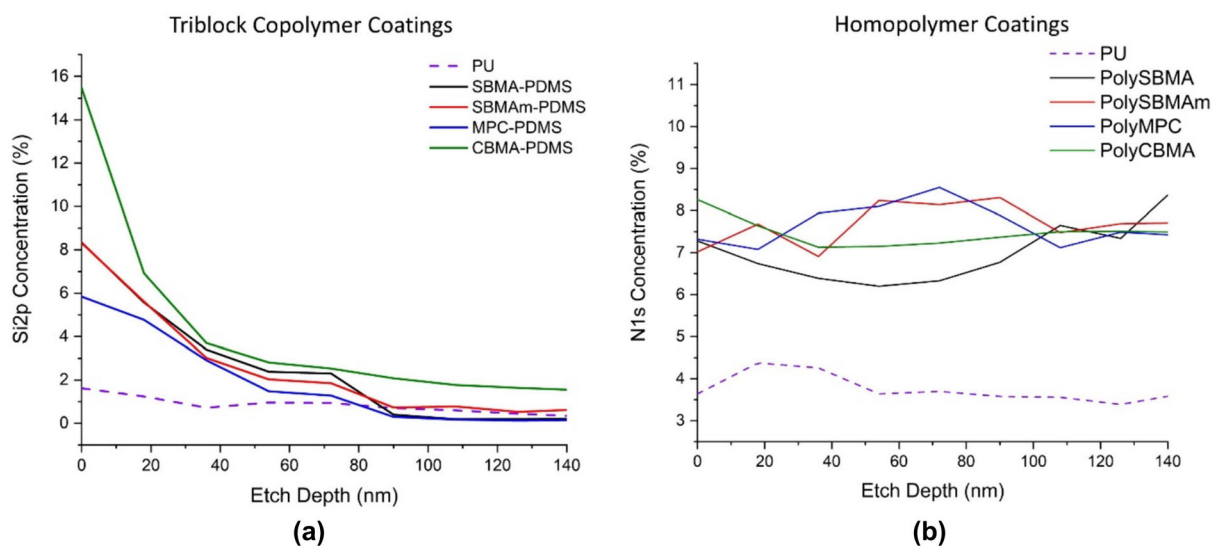


Fig. 5: (a) Si2p XPS etch depth concentration for triblock copolymer coatings and (b) N1s XPS etch depth concentration for homopolymer coatings

sideband mode, the resulting spectrum from inside and outside these domains has extremely similar intensities and peaks. The peaks seen in sideband mode are indicative of only PDMS. The C–H (1230 cm⁻¹), Si–O (1100 cm⁻¹), and Si–C (805 cm⁻¹) are the only observable peaks. This indicates that this is a very thin layer of PDMS that is present directly on the outer surface. When imaging in the “bulk” or direct mode, there is a clear difference between the spectra when taken inside a domain versus outside a domain. A stronger polyurethane signal can be seen outside the domain (1740 cm⁻¹). Inside the domain, the expected peaks of

the zwitterions and PDMS are seen which are shown in the FTIR of the polymer/additive. This result shows that the zwitterionic portion of the additive is slightly “buried” below a very thin layer of PDMS. However, the WCA analysis showed that the surface could rearrange into a more hydrophilic state upon contact with water. Thus, this thin PDMS layer does not inhibit the expression of the hydrophilic component on the surface. This arrangement and “burying” of zwitterionic portion of the polymer is a common trait when working with these extremely hydrophilic zwitterions when attached to a an incompitable hydrophobic

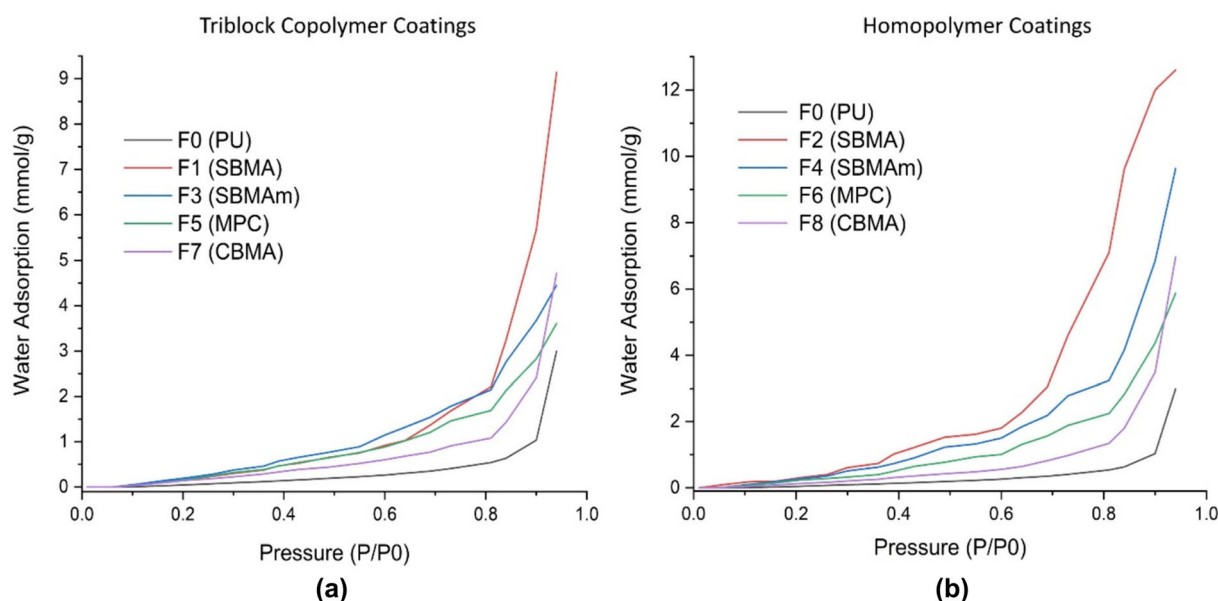


Fig. 6: Moisture adsorption measurements at 40°C for (a) triblock formulations and (b) homopolymer formulations

PDMS chain.⁵² However, it can be noted that the PDMS portion of the polymer chain is needed to help the additive/copolymer stratify to the surface.

X-ray photoelectron spectroscopy (XPS) was used in this study to quantify the elemental compositions on the surface of the coating as well as a function of depth. As shown by the contact angle, AFM, and PiFM results, the amphiphilic block copolymer additive is stratified onto the surface, and XPS confirms this result with the Si2p elemental concentration. The Si2p elemental concentration for triblock copolymer coatings is shown in Fig. 5a and the N1s homopolymer in Fig. 5b. The O1s concentration is shown in Fig. S22 and C1s concentration in Fig. S23. The initial Si2p concentration of the amphiphilic additives is at its highest point at the surface and starts to decrease as the etch depth is increased. Once the coating has been etched into around 140 nm, the concentration of the additive is not much higher than the control coating. These XPS results confirm the PiFM data as well as explain why the ATR-FTIR could not identify the presence of the additive because the concentration of the amphiphilic copolymer additive drops off before the infrared penetration depth. The XPS results for the triblock polymers also confirm the stratification of the amphiphilic polymer at the surface. As for the homopolymer coatings, the N1s concentration shows the presence of the zwitterion at the surface. However, there is not a decreasing concentration trend as the coating is etched into compared to the triblock formulations, which indicates that the homopolymers are more uniformly distributed in the polyurethane. These results confirm that the PDMS block of the polymer is important for stratification of the additives to the surface.

Moisture vapor adsorption experiments were conducted to determine which zwitterion had the best

ability to create a hydration layer at the surface. All formulations were tested in comparison with the polyurethane control to determine if there were any differences in the coatings' ability to adsorb moisture on the surface. The moisture adsorption measurements for the triblock formulations are shown in Fig. 6a and the homopolymer formulations in Fig. 6b. A few observations can be seen from these experiments. The first observation is that with the addition of the zwitterionic polymers, the moisture adsorption of the coatings increases when compared to the polyurethane control, indicating that the additives are affecting the hydrophilicity of the coatings. The second observation is that the amphiphilic coatings with PDMS have a lower moisture adsorption than the homopolymer zwitterionic coatings. This is expected as without the PDMS, these zwitterions are extremely hydrophilic, as well as there being a higher amount of the zwitterion content in these coatings due to the absence of PDMS. The third observation is that at atmospheric pressure (1.0 P/P0) for the triblock coatings, the SBMA zwitterion appears to have the highest moisture adsorption followed by CBMA, SBMAm, and MPC. These results are consistent with the previous literature in measuring the hydrophilicity of these zwitterions.^{53–55} It can be concluded that the addition of the zwitterions allows the coating to create a better hydration layer at the surface due to a higher water adsorption concentration and can be used to explain the AF/FR results. The full absorption and desorption curves for each coating are found in Figs. S24, S25, and S26. It can be noted that the control formulation has very little hysteresis between the adsorption and desorption curves when compared to the larger hysteresis of the experimental formulations at higher pressure (P/P0). However, this is little difference between the hysteresis of the

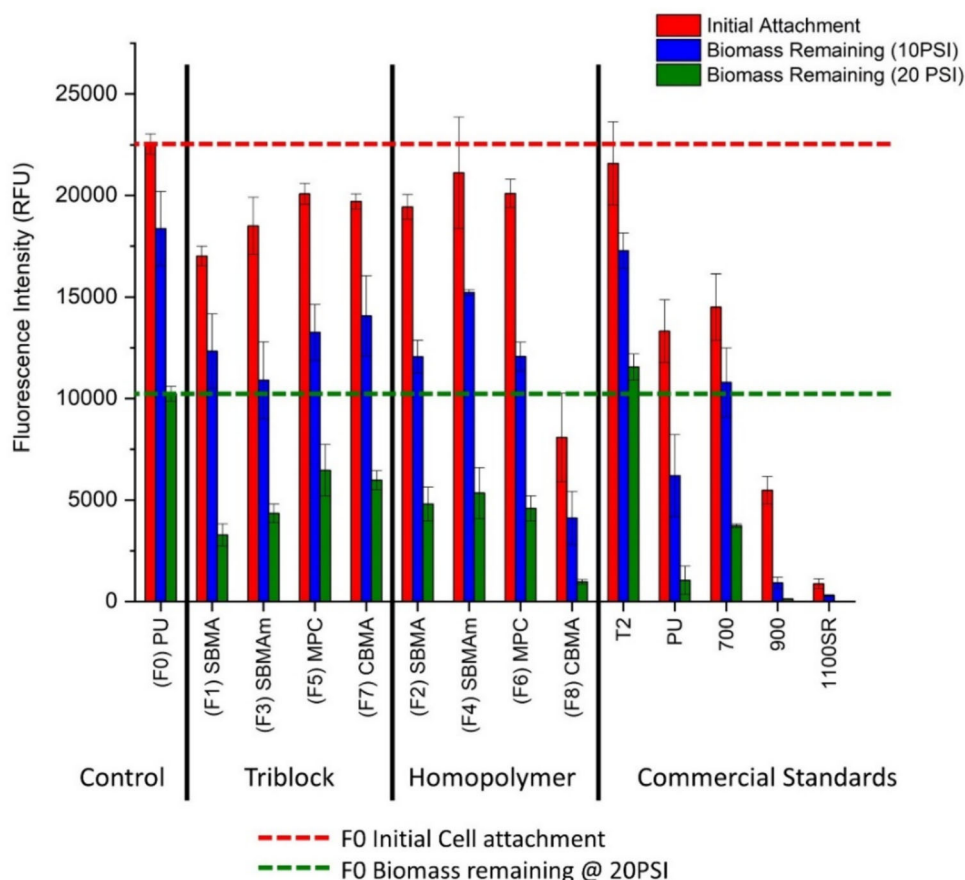


Fig. 7: Fouling-release data for *N. incerta*: total biofilm growth (red bar) and release data—biofilm remaining after 10 PSI waterjet (blue bar) and biofilm remaining after 20 PSI water jet (green bar). The dotted red line shows the initial biofilm attachment for F0/control, and the green line shows the remaining biomass remaining at 20 PSI waterjet for F0/control coating (Color figure online)

sorption curves when compared among the eight experimental formulations. This could be attributed to the water binding more strongly to the zwitterionic-based coatings when undergoing desorption.

AF/FR performance assessments via biological laboratory assays

Biological laboratory assays were performed to assess the AF/FR properties using a wide range of marine fouling organisms having differing adhesion preferences. All these assessments were carried out following 28 days of water leaching to confirm that any toxic impurities would be removed and not intervene with the results. The coatings were evaluated for leachate toxicity for *N. incerta* and *C. lytica* using procedures described elsewhere.^{44,56} The toxicity assessments are found in Figs. S27 and S28. It was found that after the leaching, the coatings were not toxic and could be used to obtain accurate AF/FR experiments.

***N. incerta* laboratory biological assays**

The diatom *Navicula incerta* is a microalga that forms a slime on the surface and tends to adhere strongly to hydrophobic surfaces rather than hydrophilic ones.^{57–59} The extent to which *N. incerta* fouled the surface varied among the commercial, experimental, and control coatings (Fig. 7—red bars). Overall, the experimental coatings had similar initial biofouling settlement, while the control polyurethane had a higher initial biofouling. The commercial amphiphilic fouling-release coatings (900 and 1100SR) had the lowest initial fouling, believed to be due to their reported amphiphilic nature. The removal of *N. incerta* was measured at two pressure levels, and the remaining biomass was reported at 10 PSI (Fig. 7—blue bars) and 20 PSI (Fig. 7—green bars). The remaining biomass of the diatom biofilm was visibly lower at 20 PSI compared to 10 PSI; however, the trends at both levels were comparable across all coatings. The addition of the zwitterionic additives increased the removal amount and decreased the amount of biofouling; this shows that the zwitterionic polymers are increasing the

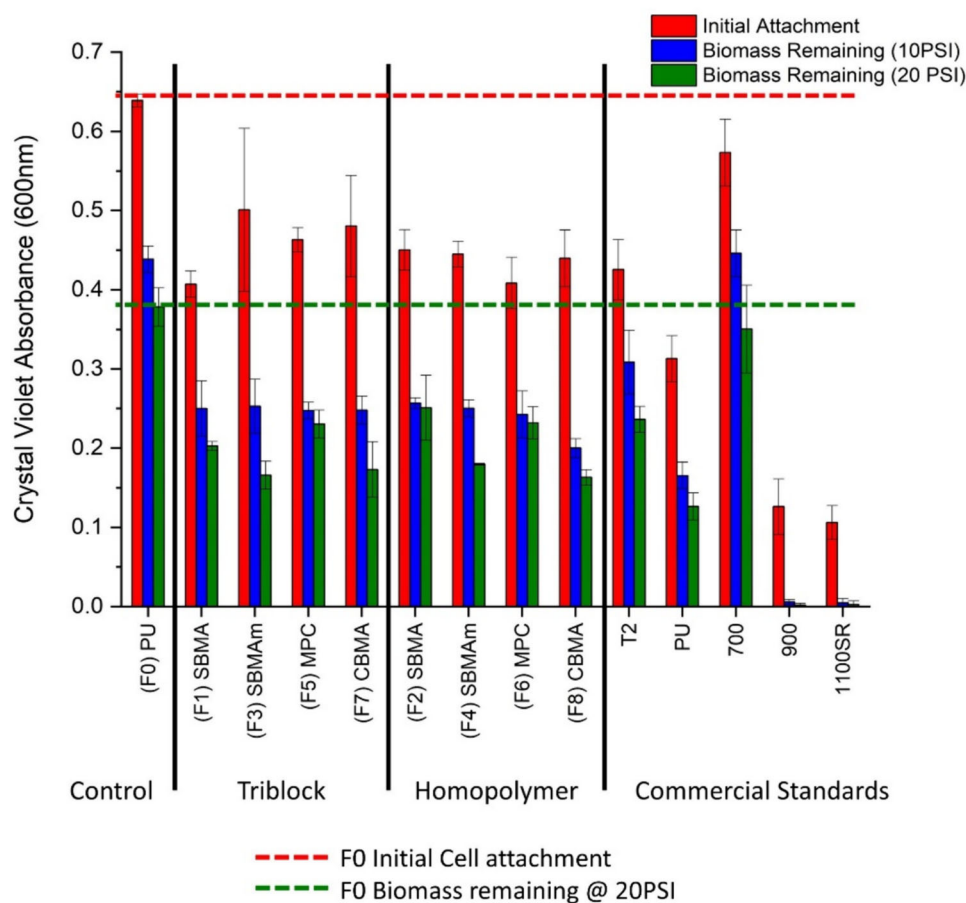


Fig. 8: Fouling-release data for *C. lytica*: total biofilm growth (red bar) and release data—biofilm remaining after 10 PSI waterjet (blue bar) and biofilm remaining after 20 PSI water jet (green bar). The dotted red line shows the initial biofilm attachment for F0/control, and the green line shows the biomass remaining at 20 PSI waterjet for F0/control coating (Color figure online)

antifouling and fouling-release properties of the control polyurethane. However, there is not as clear of a trend as to which zwitterion and type of polymer (triblock or homopolymer) improved the results the best, but some trends can be determined. All the experimental coatings were statistically significant at 0.05 level compared to the control F0 (Table S1). It can be noted that the amphiphilic SBMA and homopolymer CBMA coatings were both significantly different than the other experimental coatings. These results are even getting close to matching (as they are not significantly different) the best-performing Intersleek 1100SR commercial coating. It can be concluded that the addition of the zwitterions helps improve the AF/FR results, while the SBMA and CBMA zwitterions appear to perform the best against *N. incerta*.

C. lytica laboratory biological assays

Cellulophaga lytica is a marine bacterium that is another common biofouling organism that has been known to adhere to a wide range of hydrophobic or

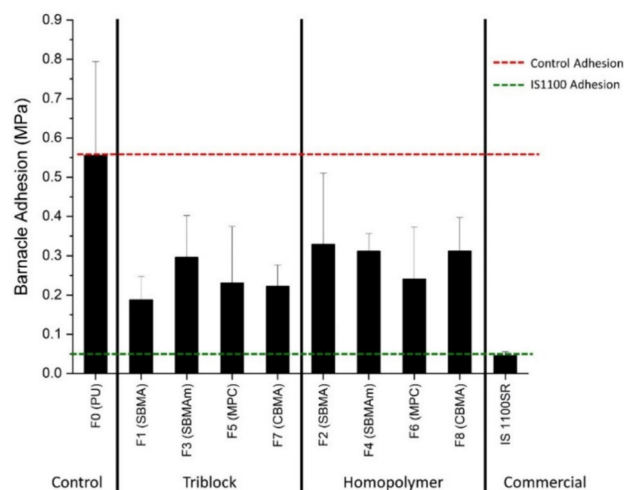


Fig. 9: Average reattached barnacle adhesion. The dotted red line shows the average adhesion for F0/control, and the dotted green line shows the average adhesion for commercial FR standard Intersleek 1100 SR (Color figure online)

hydrophilic surfaces. This organism is a precursor for harder-fouling organisms like tubeworms.³ The extent of biofouling amassed on the surface varied between the control, experimental coatings, and commercial controls (Fig. 8—red bars). Overall, the experimental coatings had a lower initial biofouling attachment than the control F0. The commercial amphiphilic fouling-release controls (900 and 1100SR) had the lowest initial fouling settlement. The fouling-release experiments were carried out at two pressure levels, and biomass remaining was reported at 10 PSI (Fig. 8—blue bars) and 20 PSI (Fig. 8—green bars). The release of the biofilm at 10 PSI was similar to 20 PSI, and the trends were alike between these two pressure levels. The addition of the zwitterionic polymers, again, increased the antifouling properties of the control polyurethane coating. At 20 PSI, the addition of these zwitterionic polymers increased the fouling-release ability as well. The experimental coatings were all statistically significant at 0.05 level compared to the control F0 coating (Table S2). This shows that, again, the zwitterionic coatings are improving the AF/FR performance. When comparing the zwitterions against each other, some trends can be observed. The homopolymer CBMA again appears to perform very well. Most of the zwitterionic coatings are not significantly different than each other, but it appears that the CBMA and SBMA zwitterions outperform the other zwitterions.

A. amphitrite laboratory biological assays

The barnacle *Amphibalanus amphitrite* is a major source of biofouling and is one of the main foulants that leads to increased fuel consumption due to heavy calcareous fouling.^{60–62} *A. amphitrite* barnacles prefer to settle on hydrophilic surfaces but can still easily settle on hydrophobic surfaces as well.^{46,47,63,64} All six barnacles used for each formulation did not exhibit any shell breakage when pushed off because the adhesive strength was less than the strength of the barnacle shell. This is an important indication that the coatings are useful in fouling-release applications. The attached barnacle adhesive strength is shown in Fig. 9. The improved fouling-release ability of the zwitterion polymers from the control F0 coating is evident. The statistical analysis is shown in Table S3. Only amphiphilic SBMA (F1) and IS1100SR coatings were significantly different than the F0 control coating. However, some other trends can be determined from this data. The amphiphilic zwitterion polymers appear to have improved FR abilities for barnacles when compared to the homopolymer coatings. The SBMA and CBMA zwitterions, again, appear to be performing the best for this organism, like the other two fouling organisms. Overall, the addition of the zwitterionic polymers increased the FR ability, and the SBMA + CBMA zwitterions improved the FR abilities the most.

Conclusions

A series of novel amphiphilic block copolymers based on hydrophobic PDMS and hydrophilic zwitterion monomers were successfully synthesized. Four different zwitterion monomers were also individually synthesized into block copolymers. Commercially available zwitterionic monomers SBMA, SBMAm, MPC, and CBMA were incorporated into the triblock copolymers as well as used in the synthesis of homopolymers. These polymers were prepared via a controlled free radical polymerization, ARGET-ATRP, that allowed for the incorporation of incompatible molecules onto a polymer chain with controlled molecular weight that could be easily incorporated into many coating systems. The polymer molecular weights were kept constant with the PDMS block being 1000 $\bar{M}_w$ and the zwitterionic blocks being 1000 $\bar{M}_w$ each: a total of 3000 $\bar{M}_w$ for the triblock polymer. The polymers/additives were incorporated into a polyurethane system at a controlled concentration (5 wt.%) to attain an amphiphilic surface that will tune the fouling-release and antifouling properties of the base PU system. This study was designed to determine and compare how the different zwitterions incorporated into a PDMS block copolymer would impact the surface properties. Through a detailed analysis of the surface composition and morphology, it was determined that polymers were self-stratified on the surface; however, the amphiphilic-based PDMS-zwitterion triblock copolymers appeared to be more amenable to being dispersed in the coating formulation and stratifying to the surface due to the presence of the PDMS block. The AF/FR laboratory assay results were greatly and significantly improved with the addition of the zwitterionic polymers. For example, the settlement of the marine bacterium *C. lytica* was reduced by 35%, and the removal improved by 45%. Barnacle adhesion reduced from 0.58 MPa for the control to 0.19 MPa for the coating with the PDMS-SBMA triblock and to 0.22 MPa for the coating with the PDMS-CBMA triblock. Although not always significantly different from each other, the SBMA and CBMA zwitterionic polymers appeared to improve the AF/FR performance the best compared to the SBMAm and MPC zwitterions. These results correlate well with moisture adsorption experiments in showing the zwitterions that had the highest water adsorption showed the most improved AF and FR properties. Overall, the surface characterization verified a self-stratified copolymer with improved AF/FR when compared to the base coating. The results of this study show that zwitterions greatly improved the AF/FR performance, with SBMA and CBMA appearing to be the ideal zwitterions to be incorporated into future coatings.

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Author contribution Joseph Dahlgren helped in conceptualization, methodology, validation, investigation, formal analysis, and writing—original draft. Shane J. Stafslie helped in methodology, investigation, and validation. Lyndsi Vanderwal worked in investigation and validation. James Bahr worked in investigation and validation. Dean C. Webster worked in supervision, funding acquisition, and writing—review and editing.

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